
CAGT

**Recall AI
Vision Document**

Version 1.0

Revision History

Date	Version	Description	Author
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Vision (Small Project)

1. Introduction

The purpose of this document is to collect, analyze, and define high-level needs and features of the **Recall AI**. It focuses on the capabilities needed by the stakeholders and the target users, and **why** these needs exist. The details of how the **Recall AI** fulfills these needs are detailed in the use-case and supplementary specifications.

The introduction of the **Vision** document provides an overview of the entire document. It includes the purpose and references of this **Vision** document.

2. Positioning

2.1 Problem Statement

The problem of	passive learning, lack of objective self-assessment, unstructured cramming before exams, and the inability to trace knowledge gaps to their root prerequisite concepts
affects	1st and 2nd-year university students in engineering and natural sciences (especially for subjects with heavy theory and hierarchical concept structures)
the impact of which is	low memorization efficiency, the illusion of competence (confusing familiarity with true mastery), ineffective revision, and repeating mistakes without addressing the foundational knowledge gaps
a successful solution would be	an AI-powered active revision assistant that automatically generates structured study plans from learning materials, conducts multi-turn interviews to objectively verify true understanding, and uses a concept graph to trace and remediate foundational knowledge gaps

2.2 Product Position Statement

For	1st and 2nd-year university students in engineering and natural sciences
Who	need to effectively revise subjects with heavy theory and hierarchical concepts, but struggle with passive learning and identifying foundational knowledge gaps
The Recall AI	is an AI-powered active revision assistant (web application)
That	automatically generates structured study plans and conducts multi-turn interviews to objectively verify true understanding based on user-uploaded materials
Unlike	existing AI study tools like Google NotebookLM, Quizlet, and RemNote
Our product	automatically traces incorrect answers back to their root prerequisite concepts using a Concept Graph Engine, injecting foundational remediation directly into the study schedule

3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Development Team (Group CAGT)	The team of 5 software engineering students responsible for building the product.	Designs, develops, tests, and deploys the Recall AI application. Ensures the system is robust, meets the defined requirements, and is completed within the 10-week project schedule.
Course Instructor / Evaluator	The lecturer and assessor for the "Introduction to Software Engineering" course.	Monitors project progress, reviews the proposed architecture, evaluates technical complexity (e.g., the Concept Graph Engine), and grades the final product based on course criteria.
AI Service Provider (Google Gemini)	The external provider of the Large Language Model APIs used in the project.	Ensures the AI infrastructure is available, responsive, and capable of returning structured JSON outputs required for the AI Study Planner and AI Examiner.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
University Student (Primary User)	1st and 2nd-year university students in engineering and natural sciences studying subjects with heavy theory and hierarchical concepts.	Uploads study materials, sets revision deadlines, participates in multi-turn AI interviews, and tracks their learning progress and knowledge gaps via the concept graph dashboard.	The Development Team (specifically the Project Manager and UI/UX Designer) is responsible for representing the users' interests to ensure the product is intuitive and solves their pain points.

3.3 User Environment

- The system is designed for individual use. Only one person (the student) is involved in completing the study and revision tasks. This is not expected to change, as the core value of the product is providing a personalized, self-paced assessment and remediation plan based on the individual's knowledge gaps.
- A typical task cycle revolves around a "Focus Session" followed by a "Verify (Interview)" session. By default, the system uses the Pomodoro technique for study sessions, implying a cycle of about 25 minutes of focused studying. The subsequent AI interview is designed to be concise, strictly bounded by a state machine to a maximum number of question-answer turns per concept (e.g., 3 turns) to control time and API costs.
- Because the core features (AI Study Planner, AI Examiner, and Concept Graph Engine) rely on an external LLM service (Google Gemini API), the application requires a stable internet connection to function. It cannot be used offline (e.g., in-flight without Wi-Fi).
- The product is being developed as a web application (a Single Page Application built with React and Vite). Therefore, users will interact with the system primarily through modern web browsers on laptops, desktops, or tablets. There are no plans for native mobile applications in the MVP scope.
- Students currently use various applications to create or read study materials, such as MS Word, PDF readers, Notion, or image viewers. Recall AI does not need direct software integration with these

applications. Instead, it accepts their outputs (pasted text, uploaded PDFs, or photographed images) as input for the Ingest & Map phase.

3.4 Alternatives and Competition

Currently, students often rely on the status quo (passive reading, highlighting, and unstructured cramming) or use existing AI-based educational tools. Below is an analysis of the primary competitive alternatives:

- **Google NotebookLM**
 - + Strengths: Excellent at generating questions/flashcards from user-uploaded materials.
 - + Weaknesses: Spaced repetition is only partial (filters "previously wrong" answers without a full spacing algorithm). Conversational quizzing is very new and its depth is unclear. Requires custom prompts for application/transfer questions. **Lacks prerequisite trace-back.**
- **Quizlet (Q-Chat, Learn Mode)**
 - + Strengths: Mature spaced repetition algorithm (since 2017). Built-in application questions ("Apply My Knowledge") and multi-turn stateful quizzing options.
 - + Weaknesses: Focuses mostly on independent terms rather than connected concepts. **Lacks prerequisite trace-back.** Not specifically optimized for Vietnamese engineering terminology.
- **RemNote**
 - + Strengths: Advanced 3-axis spaced repetition model (recall, stability, difficulty). Great for generating cards from notes.
 - + Weaknesses: Does not have built-in application questions or multi-turn conversational quizzing. **Lacks prerequisite trace-back.**
- **Custom GPT / Claude Project (Homegrown Alternative)**
 - + Strengths: Highly flexible, capable of generating questions and holding multi-turn conversations.
 - + Weaknesses: High friction for users (requires manual prompt engineering and repasting materials). No built-in spaced repetition system (users must manually manage their own review schedules). The user has to manage the conversational flow. **Lacks automated prerequisite trace-back.**

Primary Differentiator: None of the competitors currently offer an automated "Concept Graph Engine" that traces an incorrect answer back to its root prerequisite concepts from freely uploaded user materials.

4. Product Features

No.	Feature	Description	Priority
1	AI Study Planner (Ingest & Map)	Processes user-uploaded materials (text/images/PDFs) via LLM to automatically extract a list of concepts, estimate difficulty, and build a prerequisite relationship graph. Outputs structured data for scheduling	High
2	Concept Graph Engine (Remediate)	The core algorithm that traces weak scores back through the prerequisite graph. It automatically identifies foundational knowledge gaps and injects remediation sessions into the user's schedule	High

3	AI Examiner (Verify / Interview)	Conducts stateful, multi-turn oral quizzes based strictly on the uploaded material. It distinguishes genuine understanding from rote memorization using a bounded state machine (e.g., max 3 turns per concept)	High
4	Dashboard & Graph Visualization	Display the user's study progress and upcoming deadlines. Features and interactive concept graph diagram (via <code>react-flow</code>) color-coded by mastery level, allowing users to visually pinpoint their weak areas	High
5	Focus Session	Provides a timer (defaulting to the Pomodoro technique) to track study sessions. The timer acts as a soft estimate, allowing users to manually end the session when they actually finish a task	Medium
6	Account Management	Basic user authentication (sign up/ sign in via email and password) and profile management to save individual study plans and progress	Low

5. Non-Functional Requirements

- **Platform & Environmental Requirements:**
 - + The system must be a web-based Single Page Application (SPA), accessible via modern web browsers (Chrome, Safari, Edge, Firefox).
 - + The system requires an active, stable internet connection to function, as it relies heavily on external cloud services (Google Gemini API) for its core AI operations.
- **Performance & Responsiveness:**
 - + The UI must feel responsive. Graph visualization rendering (using `react-flow`) must remain smooth for typical study documents (which generally contain fewer than 50 concepts).
 - + AI response times during the "AI Examiner" interview must be kept within acceptable conversational limits to prevent user frustration.
- **Cost & Resource Constraints:**
 - + To manage third-party API costs (Gemini API) and rate limits, the system must enforce strict boundaries on AI usage: multi-turn interviews are hard-capped (e.g., maximum 3 turns per concept), and daily request limits per user should be implemented.
- **Reliability & Fault Tolerance:**
 - + **AI Fallback Mechanism:** If the external AI service fails, times out, or exceeds quotas, the system must gracefully degrade to a self-grading mode. It will use previously generated questions as static flashcards, allowing the user to self-assess without breaking the scheduling engine.
 - + **Data Integrity:** The system must use a relational database (PostgreSQL with ACID transactions) to ensure the integrity of the concept graph (preventing orphaned concepts or corrupted prerequisite edges).
 - + **Cycle Prevention:** The system must validate that AI-generated concept graph is a Directed Acyclic Graph (DAG). Any cyclic relationships must be rejected or flagged during the save process to prevent infinite loops in the trace-back algorithm.